

Brandon Stile

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EDUCATION

University of Central Florida
Bachelor of Science: Computer Science

Orlando, FL
Expected Grad May 2029

EXPERIENCE

Embedded Software Engineer August 2025 – Present
VEX University Robotics Competition – UCF7 Daydream Orlando, FL

- Collaborated with a team of **25+** people to design, build, and program two autonomous and driver controlled robots that placed **35th out of 150** teams at the **VEX U 2026 World Championship**
- Created a synthetic data generation pipeline automatically generating **60,000+** photos for each CNN training set using **Unreal Engine** and **NVIDIA Deep Learning Dataset Synthesizer** improving CNN accuracy
- Constructed a **computer vision pipeline** to detect, classify, and retrieve game objects, achieving **95% collection success** with a CNN reaching **99.7% precision** and **96% recall** in simulation

Full Stack Intern December 2025 – March 2026
Minoria Tech Orlando, FL

- Co-led a team of **15** developers to design and build a **SaaS** business management application to provide users with a quote fulfillment pipeline, customer relations, and accounting software using **React**, **Next.js**, and **TypeScript**
- Refactored **10k+** lines of legacy code using **Next.js App Router** to cut initial **JavaScript** bundle size by **30%**
- Overhauled frontend UI components, raising **Lighthouse performance scores** from **62 to 94** and decreasing average page load time by **60%**

Full Stack Developer August 2025 – January 2026
Robotics Club of Central Florida Orlando, FL

- Designed and launched a high-performance club website portal using **Svelte** and **Tailwind CSS**, driving **1,000+** monthly visitors while introducing a new sponsorship landing page with structured tiers and clear benefits
- Oversaw user credentials and member data for **100+ active** accounts using **Prisma** database management
- Optimized website for mobile devices, which account for **68%** of total traffic, by creating dynamic viewports across the website to prevent hidden UI elements and element overflow

PROJECTS

DAVE | 1st Overall | *React, Gemini API, Next.js, TypeScript, Python* July 2026

- Developed a wearable, motion-tracking volleyball coach, earning **1st** place at BloomKnights among **86** projects and **248** participants, with a responsive frontend and 3D swing visualization reconstructed from **IMU** data
- Engineered a real-time pipeline processing **200 Hz IMU** data and **Random Forest** outputs, integrating the **Gemini 3.5 Flash API** for automated swing ratings and personalized coaching
- Built an **LLM-powered chat** interface with **Next.js** and **React**, enabling athletes to receive personalized technique adjustments and actionable recommendations using collected performance data

MiniCache | *Python, Data Structures & Algorithms* July 2026

- Built an **LRU cache** from-scratch in Python using a custom hash table, doubly linked list, and min-heap without using Python's built-in hash-table or min-heap and **O(1) average-case** get/put
- Implemented **lazy Time-To-Live** expiration with tombstoning to safely discard expired entries reconcile stale heap nodes on key updates/deletes, reducing re-indexing overhead while preserving correct eviction and expiry behavior

TECHNICAL SKILLS

Languages: Java, TypeScript, HTML, CSS, JavaScript, C/C++, Python, C#

Libraries and Frameworks: React, Next.js, Tailwind CSS, Prisma, Svelte, TensorFlow, NumPy, NDDS

Developer Tools: Git, GitHub, VS Code, Docker, Google Colab, Gemini API, Unreal Engine